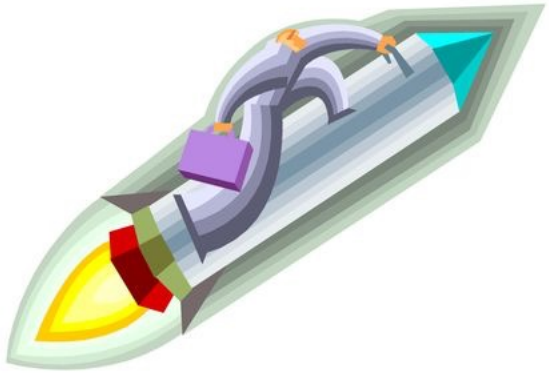
STA 2002, Summer 2024
Probability and Statistics II

Hypothesis Testing
Part I

Example: Aircrew Escape System

- **Aircrew escape systems are powered by a solid propellant. Rocket motor contains a propellant.**
- **To reject seat properly, specification require that the mean burn rate must be 50cm/s. Burning too slow or too fast are both not safe.**
- **10 samples are tested to determine whether the mean burn rate is 50**





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Outline

- Null hypothesis H_0 vs. alternative hypothesis H_1
  - Null hypothesis: initial belief/claim/assumption
  - Alternative hypothesis: competing belief/claim/assumption
- Test statistic *T* and critical region
- Type I and Type II error
- Significance level
- p-value

Reading: Chapter 8.1, 8.2, 8.3 in our textbook.

Statistical hypothesis test

- Statistical hypothesis testing of parameters are the fundamental methods used at the data analysis stage of a comparative experiment

- Many types of decision-making problems, tests, or experiments can be formulated as hypothesis testing problems

Statistical Hypothesis

A **statistical hypothesis** is a statement about the parameters of one or more populations.

Hypothesis Test

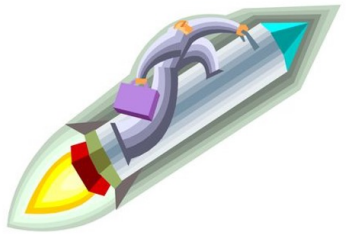
- **Interested in burning rate of a solid propellant**
- **Burning rate is a random variable**
- **Deciding whether or not the mean Burning rate is 50 cm/s**

Null hypothesis H_0: $\mu = 50$ centimeters per second

Alternative hypothesis H_1: $\mu \neq 50$ centimeters per second

- **A procedure leading to a decision about a particular hypothesis using data is called a test of a hypothesis**





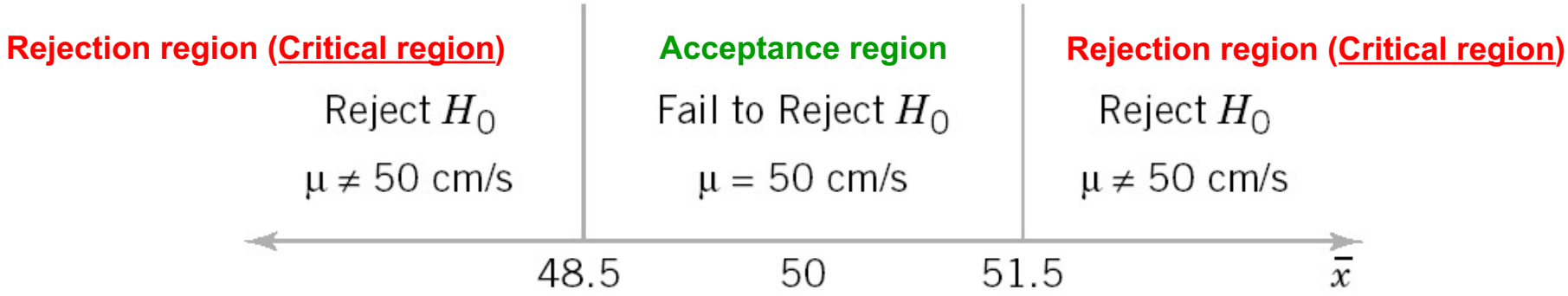
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Testing a hypothesis

- **Testing a hypothesis involves taking a random sample, computing a test statistic from the sample data, and then use the test statistic to make a decision about the null hypothesis**
- **E.g. Take 10 samples**

H_0: $\mu = 50$ centimeters per second

H_1: $\mu \neq 50$ centimeters per second



Note: By convention, we say that we reject H_0 or we fail to reject H_0.

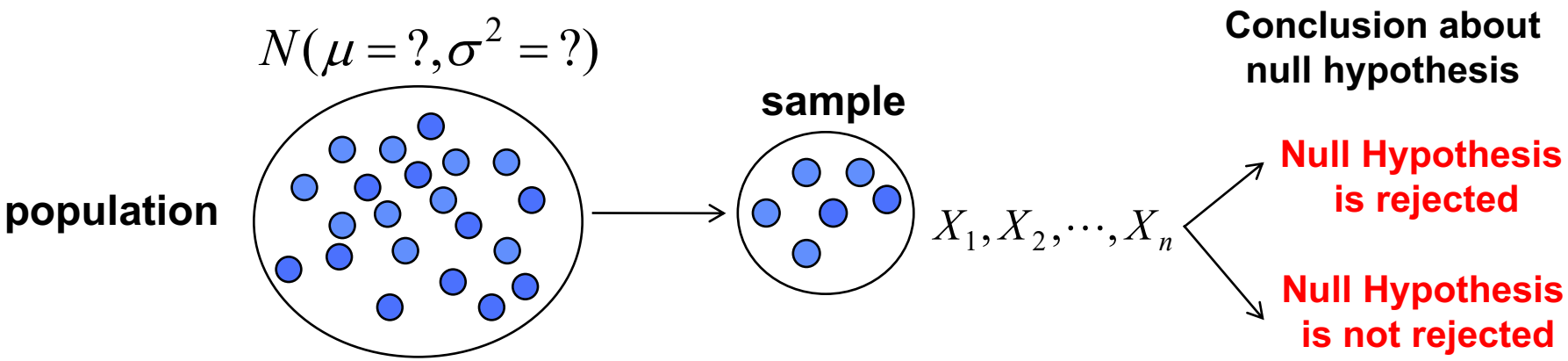
Statistical Hypothesis Testing

Hypothesis-testing procedures rely on using the information in a random sample from the population of interest.

If this information is *consistent* with the hypothesis, then we will conclude that the hypothesis is true; if this information is *inconsistent* with the hypothesis, we will conclude that the hypothesis is false.

Statistical Hypothesis Testing

- For example, comparing the mean of a population to a specified value



Null Hypothesis $\begin{cases} H_0 : \mu = \mu_0 \\ H_a : \mu \neq \mu_0 \end{cases}$ $\begin{cases} H_0 : \sigma^2 = \sigma_0^2 \\ H_a : \sigma^2 \neq \sigma_0^2 \end{cases}$

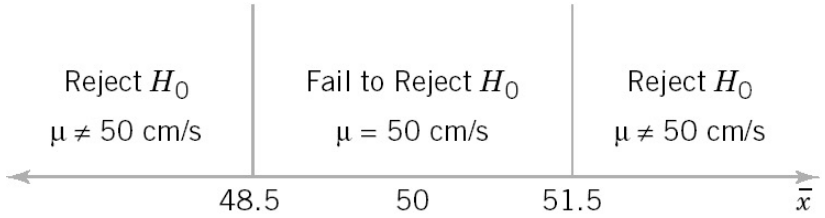
Alternative Hypothesis

Statistical Hypothesis

A **statistical hypothesis** is a statement about the parameters of one or more populations.

Key Questions

- How to set-up the hypothesis test?
- How to make decision?
  - How to choose detection statistic?
  - How to determine threshold?



- Concepts
  - Test statistic
  - Decision: “Reject” or “Fail to reject”
  - Errors: Type-I and Type-II errors
  - Significance level
  - p-value

Class Activity

(1) Is the coin fair? P(tails)=P(heads)

A. H_0: P = ½ B. H_1: P = ½ C. H_0: P ≠ ½

(2) A machine produces product (X) with mean μ, variance σ^2

Is the variability under control?

A. H_1: $\sigma^2 \leq \sigma_0^2$ B. H_0: $\sigma^2 > \sigma_0^2$ C. H_0: $\sigma^2 \leq \sigma_0^2$

(3) Do we support the hypothesis that the machine produce an item of a size larger than a known μ_0?

A. $H_1 : \mu \leq \mu_0$ B. $H_0 : \mu > \mu_0$ C. $H_0 : \mu \leq \mu_0$

Type of Hypothesis

Simple Hypothesis : only take on one value

Testing two possible values of the parameter

$H_0: \mu = 10$ **null hypothesis**

$H_1: \mu = 15$ **alternative hypothesis**

Composite Hypotheses :

One-sided Hypothesis Test:

Testing a range of values

$H_0: \mu = 10$ **null hypothesis**

$H_1: \mu > 10$ **alternative hypothesis**

$H_0: \mu = 10$

$H_1: \mu < 10$

Two-sided Hypothesis Test:

$H_0: \mu = 10$

$H_1: \mu \neq 10$

e.g. X: customers' waiting time in a bank

Errors in Hypothesis Test

Type I Error

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error.**

Probability of Type I Error

$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

Type II Error

Failing to reject the null hypothesis when it is false is defined as a **type II error.**

Probability of Type II Error

$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

Decision	H_0 Is True	H_0 Is False
Fail to reject H_0	no error	type II error
Reject H_0	type I error	no error

Example

$$\alpha = P(\bar{X} < 48.5 \text{ when } \mu = 50) + P(\bar{X} > 51.5 \text{ when } \mu = 50)$$

Type I error

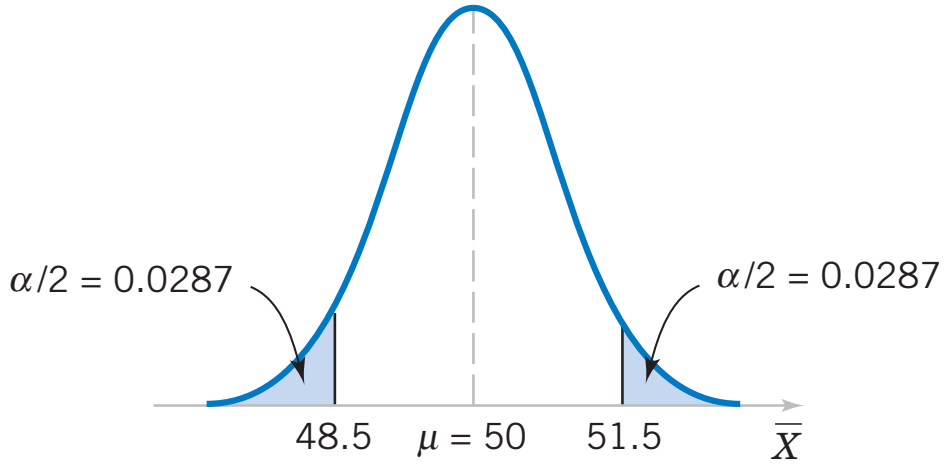
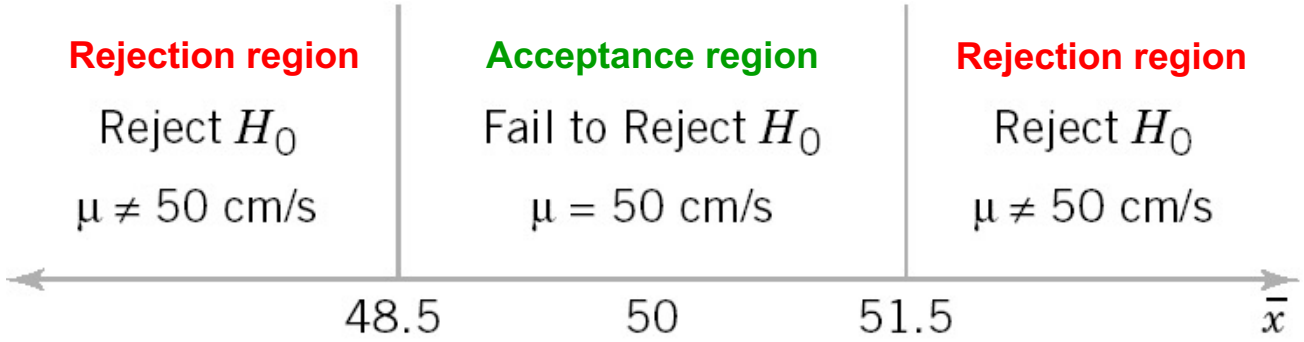


Figure 9-2 The critical region for H_0: $\mu = 50$ versus H_1: $\mu \neq 50$ and $n = 10$.



Suppose that the standard deviation of burning rate is $\sigma = 2.5$ centimeters per second

$$\alpha = P(\bar{X} < 48.5 \text{ when } \mu = 50) + P(\bar{X} > 51.5 \text{ when } \mu = 50)$$

n = 10 samples

$$\sigma/\sqrt{n} = 2.5/\sqrt{10} = 0.79.$$

$$z_1 = \frac{48.5 - 50}{0.79} = -1.90 \qquad \text{and} \qquad z_2 = \frac{51.5 - 50}{0.79} = 1.90$$

$$\alpha = P(Z < -1.90) + P(Z > 1.90) = 0.0287 + 0.0287 = 0.0574$$

$$\boldsymbol{H_1: \mu = 52}$$

$$\beta = P(48.5 \leq \bar{X} \leq 51.5 \text{ when } \mu = 52)$$

The z-values corresponding to 48.5 and 51.5 when $\mu = 52$ are

$$z_1 = \frac{48.5 - 52}{0.79} = -4.43 \qquad \text{and} \qquad z_2 = \frac{51.5 - 52}{0.79} = -0.63$$

$$\begin{aligned}\beta &= P(-4.43 \leq Z \leq -0.63) = P(Z \leq -0.63) - P(Z \leq -4.43) \\ &= 0.2643 - 0.0000 = 0.2643\end{aligned}$$

Significance Level

- **Generally, the statistician controls type-I error probability α when she selects the critical values**
- **Set type-I error probability at a desired level called significance level**
- **Deal with the corresponding type-II error β resulted from this**

> A widely used procedure in hypothesis testing is to use a type 1 error or significance level of $\alpha = 0.05$. This value has evolved through experience, and may not be appropriate for all situations.

Typical value for significance level:
0.1, 0.05, 0.001

p-value

- Fixed significance level report the results of a hypothesis test
- Significance level is an upper bound by α
- But this way we have no idea how strong the evidence is
- Also report p-value of the statistic

p-value

- The p-value is the probability that the test statistic will take on a value that is at least as extreme as the observed value of the statistic when the null hypothesis H_0 is true.
  - Given a significance level α
  - If the p-value is smaller than α $\Rightarrow$ Reject H_0
  - If the p-value is larger than α $\Rightarrow$

    there is not enough evidence to reject H_0

Example

A manufacturer is interested in the output voltage of a power supply used in a PC. Output voltage is assumed to be normally distributed, with standard deviation 0.25 Volts, and the manufacturer wishes to test $H_0: \mu = 5$ Volts against $H_1: \mu \neq 5$ Volts, using n=8 units.

(1) The acceptance region is $4.85 \leq \bar{x} \leq 5.15$. Find the value of α.

(2) Calculate the p-value if the observed statistic is

(a) $\bar{x} = 5.2$ (b) $\bar{x} = 4.7$ (c) $\bar{x} = 5.1$



Example (critical region).

$\sigma = 0.25$. $n = 8$

$H_0: \mu = 5$ v.s $H_1: \mu \neq 5$

(1) Fail to reject H_0 (Accept H_0) if

$$4.85 \leq \bar{x} \leq 5.15$$

$$\bar{X} \sim N(\mu, \tfrac{1}{n}\sigma^2)$$

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$4.85 \leq \bar{x} \leq 5.15$$

$$\Leftrightarrow \frac{4.85-\mu}{\sigma/\sqrt{n}} \leq \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \leq \frac{5.15-\mu}{\sigma/\sqrt{n}}$$

Probability of Type-1 error:

$$\alpha = P(\bar{x} \notin [4.85, 5.15] \mid H_0: \mu = 5 \text{ is true})$$

$$= 1 - P\left[\frac{4.85-5}{0.25/\sqrt{8}} \leq \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \leq \frac{5.15-5}{0.25/\sqrt{8}}\right]$$

$\downarrow$ $N(0,1)$

$$= 1 - [\Phi(1.697) - \Phi(-1.697)] = 0.0897$$

(you can also use R command pnorm(x, 0, 1) to find $\Phi(x)$)

P-value calculation

p-values. $H_1: \mu \neq 5$ is a two-sided hypothesis test

Test statistic $z_0 := \frac{\bar{x}-5}{0.25/\sqrt{8}}$

$$\text{p-value} := P(|Z| > |z_0|)$$

$Z \sim N(0,1)$

$$= 2 \cdot P(Z > |z_0|)$$

(a) $\bar{x} = 5.2 \Rightarrow z_0 = \frac{0.2}{0.25/\sqrt{8}}$

$$\text{p-value} = 2 \cdot (1 - \Phi(|z_0|)) = 2 \times \left(1 - \Phi\left(\frac{0.2}{0.25/\sqrt{8}}\right)\right) = 0.024$$

(b) $\bar{x} = 4.7 \Rightarrow z_0 = -\frac{0.3}{0.25/\sqrt{8}}$

$$\text{p-value} = 2 \times (1 - \Phi(|z_0|) = 2 \times \left(1 - \Phi\left(\frac{0.3}{0.25/\sqrt{8}}\right)\right) = 0.0007$$

(c) $\bar{x} = 5.1 \Rightarrow z_0 = \frac{0.1}{0.25/\sqrt{8}}$

$$\text{p-value} := 2 \times (1 - \Phi(|z_0|) = 2 \times \left(1 - \Phi\left(\frac{0.1}{0.25/\sqrt{8}}\right)\right) = 0.2579$$

In summary, if significance level $\alpha = 0.05$.

Then we reject H_0 in (a), (b)

fail to reject H_0 in (c).

Hypothesis Testing Procedures

1. **Parameter of interest:** From the problem context, identify the parameter of interest
2. **Null hypothesis H_0:** State the null hypothesis – initial belief/claim/assumption
3. **Alternative hypothesis H_1:** Specify an appropriate alternative hypothesis
4. **Test statistic:** Determine an appropriate test statistic
5. **Reject H_0 if :** State the rejection criteria for the null hypothesis
6. **Computation:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value
7. **Draw conclusion:** Decide whether or not H_0 should be rejected and report that in the problem context
8. **Report p-values**

- **Three ways to perform hypothesis test**
  - **Rejection region**
  - **Compare p-value with α**
  - **Using confidence interval**

Hypothesis testing for normal mean

Inference on the Mean of a Normal Population – Known Variance

Null Hypothesis $H_0 : \mu = \mu_0$

Test Statistic $Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Distribution under H_0 $Z_0 \sim N(0,1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : \mu \neq \mu_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$
$H_1 : \mu > \mu_0$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\bar{X} - z_\alpha\frac{\sigma}{\sqrt{n}}, +\infty)$
$H_1 : \mu < \mu_0$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$(-\infty, \bar{X} + z_\alpha\frac{\sigma}{\sqrt{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include μ_0

Why this gives the desired significance level?

e.g $H_0: \mu = \mu_0$

$H_1: \mu \neq \mu_0$

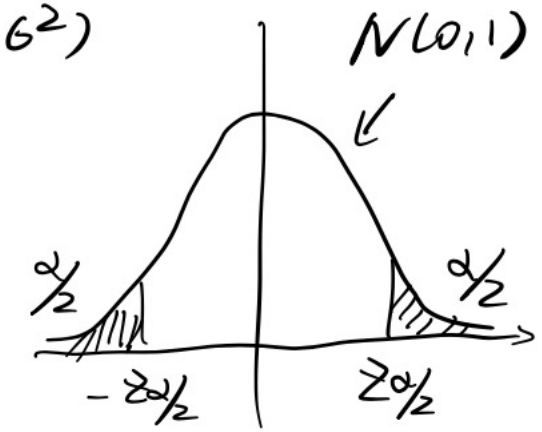
$\alpha = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

$= P(|z| > z_{\alpha/2} \mid \mu = \mu_0)$

$\downarrow$ $X_1, \ldots, X_n \sim N(\mu_0, \sigma^2)$

$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$

$= 2 \times \frac{\alpha}{2} = \alpha.$



Example: Battery life

- The life in hours of a battery is known to be approximately normally distributed with standard deviation 1.25 hours
- A random sample of 10 batteries has a mean life of 40.5 hours
- Is there an evidence to support the claim that battery life exceeds 40 hours? Use. $\alpha = 0.05$



Example: $\sigma = 1.25$, $n = 10$, $\alpha = 0.05$, $\bar{x} = 40.5$

parameter: μ

Hypothesis: $H_0: \mu = 40$ $(\mu_0 = 40)$

$H_1: \mu > 40$

Test statistic. $z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{40.5 - 40}{1.25/\sqrt{10}} = 1.265$

Reject H_0 if

① critical region approach: if $z_0 > z_\alpha = 1.645$

since $z_0 = 1.265 < 1.645$

Fail to reject.

② p-value approach.

p-value $= P(Z > z_0) = 1 - \Phi(1.265) = 0.103$

$> \alpha = 0.05$

Fail to reject.

③ C.I. approach. 95% one-sided C.I

$[\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}, +\infty) : [39.85, +\infty)$

$\mu_0 = 40 \in [39.85, +\infty)$

Fail to reject H_0.

Conclusion: We cannot reject H_0.

There is not enough evidence to support H_1.

Inference on the Mean of a Normal Population – Unknown Variance

Null Hypothesis $H_0 : \mu = \mu_0$

Test Statistic $t_0 = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Distribution under H_0 $t_0 \sim t(n-1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : \mu \neq \mu_0$	$\lvert t_0 \rvert > t_{\alpha/2}(n-1)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}]$
$H_1 : \mu > \mu_0$	$t_0 > t_{\alpha}(n-1)$	$\mathrm{P}(T > t_0)$	$[\bar{X} - t_{\alpha}(n-1)\frac{S}{\sqrt{n}}, +\infty)$
$H_1 : \mu < \mu_0$	$t_0 < -t_{\alpha}(n-1)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{X} + t_{\alpha}(n-1)\frac{S}{\sqrt{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include μ_0

Example

9-59. A 1992 article in the *Journal of the American Medical Association* ("A Critical Appraisal of 98.6 Degrees F, the Upper Limit of the Normal Body Temperature, and Other Legacies of Carl Reinhold August Wunderlich") reported body temperature, gender, and heart rate for a number of subjects. The body temperatures for 25 female subjects follow: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9, and 99.0.

(a) Test the hypothesis H_0: $\mu = 98.6$ versus H_1: $\mu \neq 98.6$, using $\alpha = 0.05$. Find the P-value.

Sample mean = 98.264
Sample variance = 0.2324





Example (unknown variance).

$n=25,\ \bar{x}=98.264,\ s^2=0.2324,\ \alpha=0.05$

parameter: mean body temperature μ

Hypothesis's $H_0: \mu = 98.6$ $(\mu_0 = 98.6)$

$H_1: \mu \neq 98.6$ (two-sided)

Test statistic $t_0 = \dfrac{\bar{x}-\mu_0}{s/\sqrt{n}} = \dfrac{98.264-98.6}{\sqrt{0.2324/25}}$

$= -3.4849$

Reject H_0 if

① critical region. $t_{\alpha/2}(n-1) = t_{0.025}(24) = 2.0639$

$|t_0| > t_{\alpha/2}(n-1)$

Reject H_0

② p-value. $p\text{-value} = 2P(t(24) > 3.4849)$

$= 0.0019 < 0.05$

Reject H_0

③ C.I. $\left[\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}\right]$

$[98.065, 98.463]$

$\mu_0 = 98.6 \notin [98.065, 98.463]$

Reject H_0

very strong evidence to reject H_0

Hypothesis testing for normal variance

Inference on the Variance of a Normal Population

Null Hypothesis $H_0 : \sigma^2 = \sigma_0^2$

Test Statistic $\chi_0^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$

Distribution under H0 $\chi_0^2 \sim \chi^2(n-1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : \sigma^2 \neq \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2}(n-1)$ *or* $\chi_0^2 < \chi^2_{1-\alpha/2}(n-1)$	Approximation $2\min\{P(\chi^2 > \chi_0^2), P(\chi^2 < \chi_0^2)\}$	$[\dfrac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)}, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}]$
$H_1 : \sigma^2 > \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha}(n-1)$	$P(\chi^2 > \chi_0^2)$	$[\dfrac{(n-1)s^2}{\chi^2_{\alpha}(n-1)}, +\infty)$
$H_1 : \sigma^2 < \sigma_0^2$	$\chi_0^2 < \chi^2_{1-\alpha}(n-1)$	$P(\chi^2 < \chi_0^2)$	$[0, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include σ_0^2

Example

An automatic filling machine is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounces)^2. If the variance of fill volume exceeds 0.01 (fluid ounces)^2, an unacceptable proportion of bottles will be underfilled or overfilled. Is there evidence in the sample data to suggest that the manufacturer has a problem with underfilled or overfilled bottles? Use $\alpha = 0.05$, and assume that fill volume has a normal distribution.



Example: $S^2 = 0.0153$. $n = 20$, $\alpha = 0.05$

parameter: σ^2

Hypothesis $\begin{cases} H_0: & \sigma^2 = 0.01 \quad (\sigma_0^2 = 0.01) \\ H_1: & \sigma^2 > 0.01 \end{cases}$

Test statistic. $\chi_0^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{19 \times 0.0153}{0.01}$

$= 29.07$

Reject H_0 if

① critical region. $\chi_\alpha^2(19) = 30.14$

since $\chi_0^2 = 29.07 < \chi_{0.05}^2(19)$

fail to reject H_0

② p-value.

p-value $= P(\chi^2(19) > \chi_0^2) = P(\chi^2(19) > 29.07)$

$= 0.065 > \alpha = 0.05$

Fail to reject

② C.I. $\left[\frac{(n-1)S^2}{\chi_\alpha^2(n-1)}, +\infty\right)$. $[0.0096, +\infty)$

$1 \in [0.0096, +\infty)$ Fail to reject H_0

Hypothesis testing for two-samples

Hypothesis testing for equality of two normal means

Inference for Differences in Means of Two Normal Distributions (known variances)

Two independent random samples from two Normal distributions

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$
$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1)$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

e.g., $\Delta_0 = 0$ $(\mathrm{H}_0: \mu_X = \mu_Y)$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Inference for Differences in Means of Two Normal Distributions (known variances)

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Distribution under H0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}, \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\bar{X} - \bar{Y} - z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$(-\infty, \bar{X} - \bar{Y} + z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Example: Comparing two paints

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?



Example: $\sigma_X = \sigma_Y = 8$ $m = n = 10$

$\bar{x} = 121$ $\bar{y} = 112$ $\alpha = 0.05$

$$\begin{cases} H_0: \mu_X = \mu_Y & (\mu_X - \mu_Y = \Delta_0 = 0) \\ H_1: \mu_X > \mu_Y & (\mu_X - \mu_Y > \Delta_0 = 0) \end{cases}$$

Test statistic. $z_0 = \dfrac{\bar{x} - \bar{y}}{8\sqrt{\frac{1}{10} + \frac{1}{10}}} = 2.5156$

① critical region $z_\alpha = z_{0.05} = 1.645$

since $z_0 > z_\alpha$

Reject H_0

② p-value. p-value $= P(Z > z_0) = 1 - \Phi(2.5156)$

$= 0.0059 < \alpha = 0.05$

Reject H_0

③ C.I. 95% one-sided CI for $\mu_X - \mu_Y$.

$$\left[\bar{x} - \bar{y} - z_\alpha \cdot \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}},\ +\infty\right)$$

$[3.1147, +\infty)$

$0 \notin [3.1147, +\infty)$

Reject H_0

∴ We have strong evidence to reject H_0, (p-value very small)

Inference Difference in Means of Two Normal Distributions (unknown and equal variances)

pooled t-test

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

e.g., $\Delta_0 = 0$ $(H_0: \mu_X = \mu_Y)$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}}$$

Distribution under H_0

$$t_0 \sim t(n + m - 2)$$

Inference Difference in Means of Two Normal Distributions (unknown and equal variances)

pooled t-test

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}}$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

Distribution under H0

$$t_0 \sim t(n+m-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert t_0 \rvert > t_{\alpha/2}(n+m-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}, \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$t_0 > t_\alpha(n+m-2)$	$\mathrm{P}(T > t_0)$	$[\bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$t_0 < -t_\alpha(n+m-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Example: comparing catalysts

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in Table 10-1. Is there any difference between the mean yields? Use $\alpha = 0.05$, and assume equal variances.

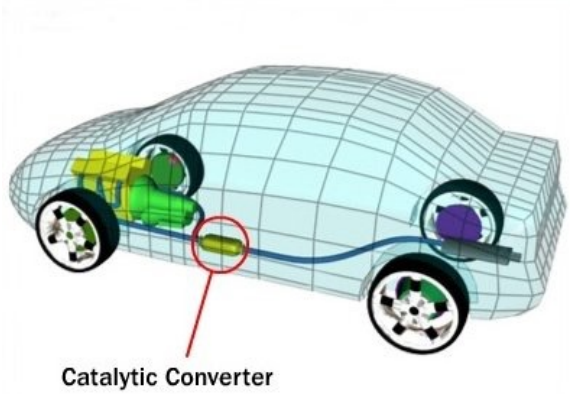
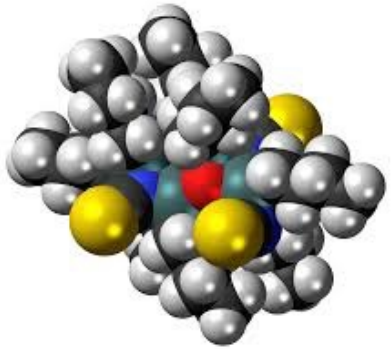


Table 10-1 Catalyst Yield Data, Example 10-5

Observation Number	Catalyst 1	Catalyst 2
1	91.50	89.19
2	94.18	90.95
3	92.18	90.46
4	95.39	93.21
5	91.79	97.19
6	89.07	97.04
7	94.72	91.07
8	89.21	92.75
	$\bar{x}_1 = 92.255$	$\bar{x}_2 = 92.733$
	$s_1 = 2.39$	$s_2 = 2.98$





Example: $\alpha = 0.05$. $m = n = 8$.

$\bar{x} = 92.255$. $S_X = 2.39$. $\bar{Y} = 92.733$. $S_Y = 2.98$

$S_p^2 = 7.2963$

$$\begin{cases} H_0: \mu_X = \mu_Y \\ H_1: \mu_X \neq \mu_Y \end{cases}$$

Test statistic. $t_0 = \dfrac{\bar{x} - \bar{Y}}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} = \dfrac{92.255 - 92.733}{\sqrt{\frac{7.2963}{4}}}$

$= -0.3539$

① critical region: $t_{\alpha/2}(14) = 2.145 > |t_0|$

fail to reject H_0

② p-value $= 2 \cdot P(t(14) > 0.3539) = 0.7287 > \alpha = 0.05$

③ C.I. $\bar{x} - \bar{Y} \pm t_{\alpha/2}(14) \cdot S_p \cdot \sqrt{\frac{1}{n} + \frac{1}{m}}$

$0 \in [-3.375, 2.419]$

Fail to Reject.

Inference Difference in Means of Two Normal Distributions (unknown and unequal variances)

Welch's t-test

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \sim t(r)$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

e.g., $\Delta_0 = 0$ $(H_0: \mu_X = \mu_Y)$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$

Distribution under H_0

$$t_0 \sim t(r)$$

Inference Difference in Means of Two Normal Distributions (unknown and unequal variances)

Welch's t-test

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor$$

Distribution under H0

$$t_0 \sim t(r)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert t_0 \rvert > t_{\alpha/2}(r)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{X} - \bar{Y} - t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}, \bar{X} - \bar{Y} + t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$t_0 > t_\alpha(r)$	$\mathrm{P}(T > t_0)$	$[\bar{X} - \bar{Y} - t_\alpha(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$t_0 < -t_\alpha(r)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{X} - \bar{Y} + t_\alpha(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Inference Difference in Means of Two Normal Distributions (paired t-test)

n independent pairs of (X, Y), but X_i, Y_i for each given *i-th* pair are dependent.

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2) \qquad Y_1, \ldots, Y_n \sim N(\mu_Y, \sigma_Y^2)$$

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n) \qquad n \text{ independent pairs}$$

$$D_i = X_i - Y_i \qquad i = 1, \ldots, n. \quad \Rightarrow \quad \frac{\overline{D} - (\mu_X - \mu_Y)}{S_D/\sqrt{n}} \sim t(n-1)$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0 \ (\mu_D = \Delta_0)$$

e.g., $\Delta_0 = 0$ $(\mathrm{H}_0: \mu_X = \mu_Y)$

Test Statistic

$$t_0 = \frac{\overline{D} - \Delta_0}{S_D/\sqrt{n}}$$

Distribution under H_0

$$t_0 \sim t(n-1)$$

Inference Difference in Means of Two Normal Distributions (paired t-test)

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$t_0 = \frac{\bar{D} - \Delta_0}{S_D/\sqrt{n}}$$

Distribution under H0

$$t_0 \sim t(n-1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert t_0 \rvert > t_{\alpha/2}(n-1)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}, \bar{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$t_0 > t_\alpha(n-1)$	$\mathrm{P}(T > t_0)$	$[\bar{D} - t_\alpha(n-1)\frac{S_D}{\sqrt{n}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$t_0 < -t_\alpha(n-1)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{D} + t_\alpha(n-1)\frac{S_D}{\sqrt{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Example

Example 8.1-5

Twenty-four girls in the 9th and 10th grades were put on an ultraheavy rope-jumping program. Someone thought that such a program would increase their speed in the 40-yard dash. Let W equal the difference in time to run the 40-yard dash—the "before-program time" minus the "after-program time." Assume that the distribution of W is approximately $N(\mu_W, \sigma_W^2)$. We shall test the null hypothesis H_0: $\mu_W = 0$ against the alternative hypothesis H_1: $\mu_W > 0$. The test statistic and the critical region that has an $\alpha = 0.05$ significance level are given by

$$t = \frac{\overline{w} - 0}{s_w/\sqrt{24}} \geq t_{0.05}(23) = 1.714.$$

The following data give the difference in time that it took each girl to run the 40-yard dash, with positive numbers indicating a faster time after the program:

0.28	0.01	0.13	0.33	−0.03	0.07	−0.18	−0.14
−0.33	0.01	0.22	0.29	−0.08	0.23	0.08	0.04
−0.30	−0.08	0.09	0.70	0.33	−0.34	0.50	0.06

Example: $n = 24$. $W = X - Y$ $\bar{W} = 0.0788$

$\begin{cases} H_0: \mu_W = 0 \\ H_1: \mu_W > 0 \end{cases}$ $S_W = 0.2549$ $\alpha = 0.05$

Test statistic. $t_0 = \dfrac{\bar{W}}{S_W/\sqrt{n}} = \dfrac{0.0788}{0.2549/\sqrt{24}}$

$= 1.514$

① critical region: $t_\alpha(n-1) = t_{0.05}(23) = 1.714$

$\because t_0 < t_\alpha(n-1)$.

fail to reject H_0

② p value $= P(t(23) > t_0) = 0.0718 > \alpha = 0.05$

③ C.I. $\left[\bar{W} - t_\alpha(n-1)\dfrac{S_W}{\sqrt{n}}, +\infty\right)$

$0 \in [-0.0104, +\infty)$

In summary. Fail to reject H_0

Hypothesis testing for equality of two normal variances

Inference on the Variance of Two Normal Populations

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{S_X^2 / \sigma_X^2}{S_Y^2 / \sigma_Y^2} \sim F(n-1, m-1)$$

$$F_{1-\alpha}(r_1, r_2) = \frac{1}{F_\alpha(r_2, r_1)}$$

Null Hypothesis

$$H_0: \sigma_X^2 = \sigma_Y^2$$

Test Statistic

$$F_0 = \frac{S_X^2}{S_Y^2}$$

Distribution under H_0

$$F_0 \sim F(n-1, m-1)$$

Inference on the Variance of Two Normal Populations

Null Hypothesis

$$H_0: \sigma_X^2 = \sigma_Y^2$$

Test Statistic

$$F_0 = \frac{S_X^2}{S_Y^2}$$

Distribution under H0

$$F_0 \sim F(n-1, m-1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach (CI for $\frac{\sigma_X^2}{\sigma_Y^2}$)
$H_1: \sigma_X^2 \neq \sigma_Y^2$	$F_0 > F_{\alpha/2}(n-1, m-1) or$ $F_0 < F_{1-\alpha/2}(n-1, m-1)$	Approximation $2\min\{P(F > F_0), P(F < F_0)\}$	$[\frac{S_X^2}{S_Y^2} F_{1-\alpha/2}(m-1, n-1), \frac{S_X^2}{S_Y^2} F_{\alpha/2}(m-1, n-1)]$
$H_1: \sigma_X^2 > \sigma_Y^2$	$F_0 > F_{\alpha}(n-1, m-1)$	$P(F > F_0)$	$[\frac{S_X^2}{S_Y^2} F_{1-\alpha}(m-1, n-1), +\infty)$
$H_1: \sigma_X^2 < \sigma_Y^2$	$F_0 < F_{1-\alpha}(n-1, m-1)$	$P(F < F_0)$	$[0, \frac{S_X^2}{S_Y^2} F_{\alpha}(m-1, n-1)]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include 1

Example: Mercedes vs. Volkswagen

10-89. A fuel-economy study was conducted for two German automobiles, Mercedes and Volkswagen. One vehicle of each brand was selected, and the mileage performance was observed for 10 tanks of fuel in each car. The data are as follows (in miles per gallon):



Mercedes		Volkswagen	
24.7	24.9	41.7	42.8
24.8	24.6	42.3	42.4
24.9	23.9	41.6	39.9
24.7	24.9	39.5	40.8
24.5	24.8	41.9	29.6



Is there enough evidence to support the claim that the variability in mileage performance is greater for a Volkswagen than for a Mercedes? $\alpha = 0.05$

Example: $n = m = 10$. $\alpha = 0.05$.

$H_0: \sigma_X^2 = \sigma_Y^2$ X: Volks

$H_1: \sigma_X^2 > \sigma_Y^2$ Y: BMW.

$S_X^2 = 15.1539$ $S_Y^2 = 0.0912$

Test statistic $F_0 = \frac{S_X^2}{S_Y^2} = 166$

① critical region. $F_\alpha(9,9) = F_{0.05}(9,9) = 3.18 < 166$

Reject H_0

② p-value $= P(F(9,9) > 166) \approx 0 \ll \alpha = 0.05$

③ C.I. $\frac{\sigma_X^2}{\sigma_Y^2} \in \left[\frac{S_X^2}{S_Y^2} F_{0.95}(9,9), +\infty \right)$

$[52.219, +\infty)$

$1 \notin [52.219, +\infty)$

Reject H_0

In summary, we have strong evidence to Reject H_0.

Example: Driving speed

Let X be the driving speed of male, and Y be the driving speed of female. Assume they are independent normal. We would like to test

$$H_0: \mu_X = \mu_Y$$

$$H_1: \mu_X > \mu_Y$$

Data.

Male (X)	Female (Y)
$n = 34$	$m = 29$
$\bar{X} = 105.5$	$\bar{Y} = 90.9$
$S_X = 20.1$	$S_Y = 12.2$

STEP 1: Decide pooled t-test or Welch's t-test

testing $\begin{cases} H_0: \sigma_X^2 = \sigma_Y^2 \\ H_1: \sigma_X^2 \neq \sigma_Y^2 \end{cases}$ $\alpha = 0.05$

Test statistic $F_0 = \frac{S_X^2}{S_Y^2} = 2.71$

Critical region. $F_{\alpha/2}(33, 28) = 2.09$

$F_{1-\alpha/2}(33, 29) = 0.49$

since $F_0 > 2.09$

Reject H_0.

$\Rightarrow$ unequal variance $\Rightarrow$ Welch's t-test

STEP 2. Welch's t-test

Test statistic. $t_0 = \dfrac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} = 3.54$

$$r = \left\lfloor \frac{\left[\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right]^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor = 56$$

$t_\alpha(56) = t_{0.05}(56) = 1.6725$

since $t_0 > 1.6725$

Reject H_0

Additional Examples

A1. Battery life

9-47. Medical researchers have developed a new artificial heart constructed primarily of titanium and plastic. The heart will last and operate almost indefinitely once it is implanted in the patient's body, but the battery pack needs to be recharged about every four hours. A random sample of 50 battery packs is selected and subjected to a life test. The average life of these batteries is 4.05 hours. Assume that battery life is normally distributed with standard deviation $\sigma = 0.2$ hour.

(a) Is there evidence to support the claim that mean battery life exceeds 4 hours? Use $\alpha = 0.05$.

(b) What is the *P*-value for the test in part (a)?

A2. Network response time

Example The response time of a distributed computer system is an important quality characteristic. The system manager wants to know whether the mean response time to a specific type of command exceeds 75 millisec. From past experience, he knows that the standard deviation of response time is 8 millisec.

If the command is executed 25 times and the response time for each trial is recorded. The sample average response time is 79.25 millisec. Formulate an appropriate hypothesis and test the hypothesis.



Solution

I. Critical region

Parameter of interest μ

Hypotheses $H_0: \mu = 75$

H1 $: \mu > 75$

$\bar{X}$ **= 79.25,** σ **= 8, n = 25**

Test statistic $Z_0 = \frac{\bar{X}-\mu 0}{\sigma/\sqrt{n}} = \frac{79.25-75}{8/\sqrt{25}}$ **=2.656**

Rejection H0 if $Z_0 > Z_\alpha = Z_{0.05}$ **= 1.645**

Reject H0. So H1 is true.

II. Confidence interval

$$H_0 : \mu = \mu_0 \quad H_1 : \mu > \mu_0 \Rightarrow \text{CI}: \left[\bar{x} - Z_\alpha \frac{\sigma}{\sqrt{n}}, +\infty\right)$$

Lower CI

$$(79.25 - 1.645 * \frac{8}{\sqrt{25}}, +\infty) = (76.62, , +\infty)$$

This DOES NOT contains 75. Reject H0. So H1 is true.

III. P-value? (exercise)

Hypothesis Testing for Proportions

Inference on a Population Proportion

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

- Using CLT, we can estimate the distribution of the sample proportion.

$$\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n} \sim N(p, \frac{p(1-p)}{n})$$

Null Hypothesis

$$H_0 : p = p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Asymptotic Distribution under H_0

$$Z_0 \sim N(0,1)$$

Inference on a Population Proportion

Null Hypothesis

$$H_0 : p = p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Asymptotic Distribution under H0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : p \neq p_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}]$
$H_1 : p > p_0$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\hat{p} - z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, 1]$
$H_1 : p < p_0$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$[0, \hat{p} + z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include p_0

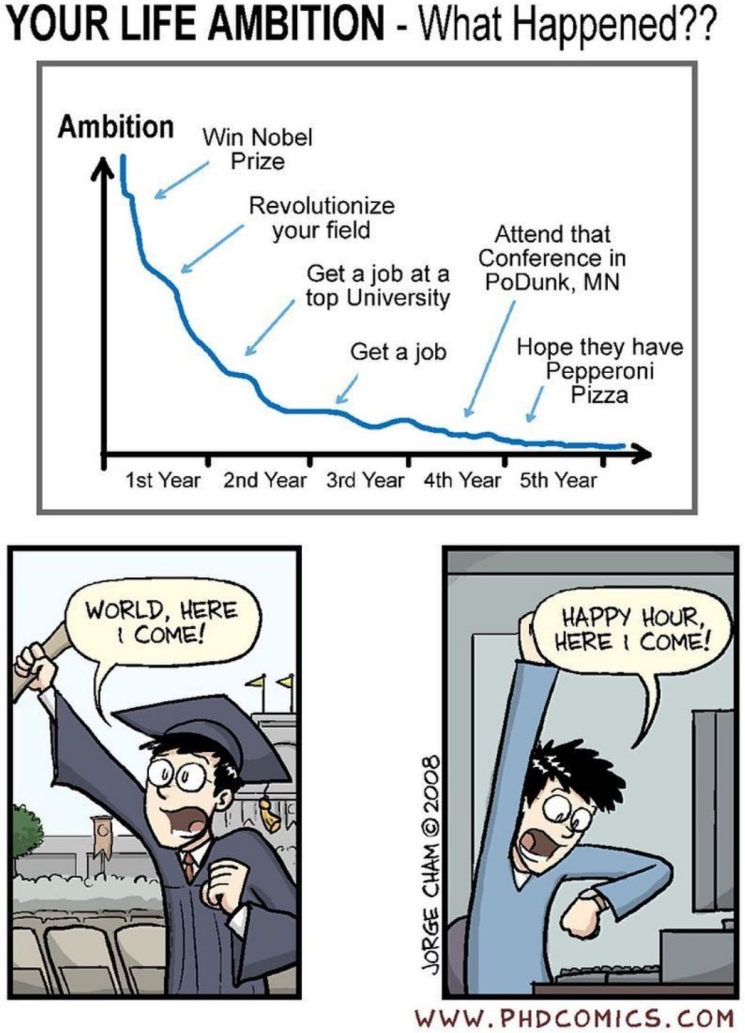
Example: Higher Education Proportion

9-54. An article in *Fortune* (September 21, 1992) claimed that nearly one-half of all engineers continue academic studies beyond the B.S. degree, ultimately receiving either an M.S. or a Ph.D. degree. Data from an article in *Engineering Horizons* (Spring 1990) indicated that 117 of 484 new engineering graduates were planning graduate study.

(a) Are the data from *Engineering Horizons* consistent with the claim reported by *Fortune*? Use $\alpha = 0.05$ in reaching your conclusions.

(b) Find the *P*-value for this test.

(c) Discuss how you could have answered the question in part (a) by constructing a two-sided confidence interval on p.



parameter of interest: p

$$\begin{cases} H_0: & p = 0.5 \\ H_1: & p \neq 0.5 \end{cases} \qquad (p_0 = 0.5)$$

Test statistic. $$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.242 - 0.5}{\sqrt{\frac{0.25}{484}}}$$

$$= -11.352$$

① critical region: $Z_{\alpha/2} = z_{0.025} = 1.96$

$|z_0| > z_{\alpha/2}$. Reject H_0

② p-value: $p = 2 \times P(z < z_0)$

$= 2 \times P(z < -11.352) = 2 \times P(z > 11.352)$

$= 7.25 \times 10^{-30} \approx 0 << \alpha$

③ C.I. $$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.242 \pm 1.96 \times \sqrt{\frac{0.242 \times (1-0.242)}{484}}$$

$0.5 \notin [0.2038, 0.2802]$ Reject H_0.

Example: Fair Die

Example 8.3-1 It was claimed that many commercially manufactured dice are not fair because the "spots" are really indentations, so that, for example, the 6-side is lighter than the 1-side. Let p equal the probability of rolling a 6 with one of these dice. To test H_0: $p = 1/6$ against the alternative hypothesis H_1: $p > 1/6$, several such dice will be rolled to yield a total of $n = 8000$ observations. Let Y equal the number of times that 6 resulted in the 8000 trials. The test statistic is

$$Z = \frac{Y/n - 1/6}{\sqrt{(1/6)(5/6)/n}} = \frac{Y/8000 - 1/6}{\sqrt{(1/6)(5/6)/8000}}.$$

If we use a significance level of $\alpha = 0.05$, the critical region is

$$z \geq z_{0.05} = 1.645.$$



Example: Fair Die

critical region approach:

The results of the experiment yielded $y = 1389$, so the calculated value of the test statistic is

$$z = \frac{1389/8000 - 1/6}{\sqrt{(1/6)(5/6)/8000}} = 1.67.$$

Since

$$z = 1.67 > 1.645,$$

the null hypothesis is rejected, and the experimental results indicate that these dice favor a 6 more than a fair die would. (You could perform your own experiment to check out other dice.)

p-value approach:

$$P(Z > 1.67) = 0.047 < 0.05$$

Reject the null hypothesis

Hypothesis testing for equality of two proportions

Inference on equality of two Population Proportions

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim Bernoulli(p_1) \\ Y_1, Y_2, \cdots, Y_{n_2} \sim Bernoulli(p_2) \end{array}\right\} \rightarrow \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \sim N(0,1)$$

For large sample sizes (CLT)

Null Hypothesis

$$H_0: p_1 = p_2$$

Test Statistic

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

$$\hat{p} = \frac{\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i}{n_1 + n_2}$$

Estimate of $p = p_1 = p_2$

Inference on equality of two Population Proportions

Null Hypothesis

$$H_0: p_1 = p_2$$

Test Statistic

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \qquad \hat{p} = \frac{\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i}{n_1 + n_2}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: p_1 \neq p_2$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\hat{p}_1 - \hat{p}_2 - z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}, \hat{p}_1 - \hat{p}_2 + z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}]$
$H_1: p_1 > p_2$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\hat{p}_1 - \hat{p}_2 - z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}, 1]$
$H_1: p_1 < p_2$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$[-1, \hat{p}_1 - \hat{p}_2 + z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}]$

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include 0

Example

8.3-13. Let p_m and p_f be the respective proportions of male and female white-crowned sparrows that return to their hatching site. Give the endpoints for a 95% confidence interval for $p_m - p_f$ if 124 out of 894 males and 70 out of 700 females returned (*The Condor*, 1992, pp. 117–133). Does your result agree with the conclusion of a test of H_0: $p_1 = p_2$ against H_1: $p_1 \neq p_2$ with $\alpha = 0.05$?

Solution

Example: $\hat{p}_1 = \hat{p}_m = \frac{124}{894}$ $\hat{p}_2 = \hat{p}_f = \frac{70}{700} = 0.1$ $\hat{p} = 0.12$

$n_1 = 894$ $n_2 = 700$ $\alpha = 0.05$

parameter of interest. $p_1 - p_2$

$$\begin{cases} H_0: & p_1 = p_2 \\ H_1: & p_1 \neq p_2 \end{cases}$$

Test statistic. $z_0 = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = 2.36$

① $z_{\alpha/2} = 1.96 < z_0$. Reject H_0.

② p-value $p = 2P(Z > z_0) = 0.018 < \alpha = 0.05$

③ C.I $\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

$0 \notin [0.007, 0.070)$ Reject H_0

Example

In a sample of $n = 200$, we have $y_1 = 22$ low birth weight babies in Country 1. In a sample of $m = 100$ we find $y_2 = 16$ low birth weight babies in Country 2. Our goal is to carry out testing on $H_0 : p_1 = p_2$ against $H_1 : p_1 \neq p_2$.

We compute

$$t = \frac{\frac{22}{200} - \frac{16}{100}}{\sqrt{\frac{38}{300}\left(1 - \frac{38}{300}\right)\left(\frac{1}{200} + \frac{1}{100}\right)}} = -1.23.$$

critical region approach: Since $|t| = 1.23 < z_{0.025} = 1.96$, we fail to reject H_0 at 5% significance level.

Additional Examples

A1. Engine controller

> A semiconductor manufacturer produces controllers used in automobile engine applications. The customer requires that the process fallout or fraction defective at a critical manufacturing step not exceed 0.05 and that the manufacturer demonstrate process capability at this level of quality using $\alpha = 0.05$. The semiconductor manufacturer takes a random sample of 200 devices and finds that four of them are defective. Can the manufacturer demonstrate process capability for the customer?

A1. $n = 200$. $\hat{p} = \frac{4}{200} = 0.02$ $\alpha = 0.05$

$$\begin{cases} H_0: p = 0.05 \\ H_1: p < 0.05 \end{cases}$$

Test statistic $z_0 = \dfrac{\hat{p} - 0.05}{\sqrt{\frac{0.05 \times (1-0.05)}{n}}} = -1.95$

① $z_\alpha = 1.645$ $|z_0| > z_\alpha$ Reject H_0.

② p-value $= P(z < z_0) = 0.0256 < \alpha$

③ C.I $\left[0, \hat{p} + z_\alpha \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right]$

$0.05 \notin [0, 0.036]$

Reject H_0.

A2. Depression

Extracts of St. John's Wort are widely used to treat depression. An article in the April 18, 2001 issue of the *Journal of the American Medical Association* ("Effectiveness of St. John's Wort on Major Depression: A Randomized Controlled Trial") compared the efficacy of a standard extract of St. John's Wort with a placebo in 200 outpatients diagnosed with major depression. Patients were randomly assigned to two groups; one group received the St. John's Wort, and the other received the placebo. After eight weeks, 19 of the placebo-treated patients showed improvement, whereas 27 of those treated with St. John's Wort improved. Is there any reason to believe that St. John's Wort is effective in treating major depression? Use $\alpha = 0.05$.

A2: $n_1 = n_2 = 100$. $\hat{p}_1 = 0.27$ $\hat{p}_2 = 0.19$ $\alpha = 0.05$

$$\begin{cases} H_0: & p_1 = p_2 \\ H_1: & p_1 \neq p_2 \end{cases}$$

Test statistic: $z_0 = \dfrac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$

$$= \frac{0.27 - 0.19}{\sqrt{0.23 \times 0.77 \times \frac{1}{50}}} = 1.35$$

① critical region. $z_{\alpha/2} = 1.64 > z_0$

fail to reject H_0

② p-value $= 2 \cdot P(Z > z_0) = 0.177 > \alpha = 0.05$

③ C.I $\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} \sqrt{\dfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \dfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

$0 \in [-0.036, 0.196]$

fail to reject H_0

Revisit: Hypothesis Testing Procedures

1. **Parameter of interest:** From the problem context, identify the parameter of interest
2. **Null hypothesis H_0:** State the null hypothesis – initial belief/claim/assumption
3. **Alternative hypothesis H_1:** Specify an appropriate alternative hypothesis
4. **Test statistic:** Determine an appropriate test statistic
5. **Reject H_0 if :** State the rejection criteria for the null hypothesis
6. **Computation:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value
7. **Draw conclusion:** Decide whether or not H_0 should be rejected and report that in the problem context
8. **Report p-values**

- **Three ways to perform hypothesis test**
  - **— Rejection region**
  - **— Compare p-value with α**
  - **— Using confidence interval**

Summary

- 3 ways to perform hypothesis test
  - Direct method: define rejection region
  - Using confidence interval
  - Compute p-value

- Several parameters of interest
  - Mean: when variance is known and unknown
  - Variance
  - Proportion
  - Comparing two populations